

Crime Scene VR — iPad Tablet Systems: Technical Report

Project: Crime Scene (Unity 6000.4.9f1, URP 17.4, BNG VR Interaction Framework, OpenXR / Meta Quest 3S)

Branch: TabletCameraAndGalleryAppsUpdate

Date: 2026-07-07

Scope: The in-game iPad tablet — architecture, apps, tooling, bugs fixed, open risks, and roadmap.

1. Executive Summary

The tablet is the player's primary interface for the investigation: it photographs evidence, reviews the photo gallery, plays recorded witness statements (audio and video), takes notes, and exposes the experience settings. Over this development cycle the tablet went from a partially-integrated collection of scripts — several of which were disconnected by a bad cherry-pick — to a **self-contained system that ships entirely inside one prefab** (`Tablet_manager.prefab`), with editor tooling that rebuilds its generated UI on demand and zero per-scene wiring.

Eighteen distinct defects were diagnosed and fixed, ranging from build-configuration failures (the APK launching as a flat black window) to subtle VR-only geometry bugs (evidence detection silently blocked by the player's own tablet). The architecture is now consistent and reproducible, but several latent risks remain — most notably unbounded photo memory growth and unrouted statement audio — and are catalogued here with suggested fixes.

2. Architecture Overview

2.1 The prefab-variant model

```
Tablet.prefab (BASE – screens, ScreenManager, home icons)
├─ Tablet_manager.prefab (VARIANT – everything else)
│   ├── ScreenManager          (inherited; +statementsScreen)
│   ├── TabletPersist          DontDestroyOnLoad across scenes
│   ├── TabletRecallInScene    "A" button summons the tablet
│   ├── PhotoLibrary          in-memory photo store (session)
│   ├── EvidenceGradingManager found-evidence registry → grade
│   ├── SettingsManager       settings.json read/write
│   ├── SettingsApplier       applies settings to each scene
│   ├── StatementLibrary      witness statements (audio+video)
│   └─ Canvas
│       ├── HomeScreen        (icons: Camera/Gallery/Notebook/Settings)
│       ├── CameraScreen      (viewfinder, shutter → IpadCamera)
│       ├── EvidenceScreen    (gallery: PhotoGalleryUI)
│       ├── NotebookScreen    (+ "Recorded Statements" link)
│       ├── SettingsScreen    (generated by builder)
│       └─ StatementsScreen   (generated by builder, runtime list UI)
```

The base prefab owns navigation and screen containers; the variant adds all game systems. Because every manager and app lives on the prefab, **both production scenes (CSHouse, CS_Outside) get the full tablet automatically**, and cross-scene persistence needs no scene setup.

2.2 Core design principles that emerged

- **Zero per-scene wiring.** Scene-dependent references (lights, the player rig's locomotion) are *auto-resolved at runtime on every scene load* rather than serialized, because a persistent prefab cannot hold scene references. `SettingsApplier` and the gaze-door system both follow this rule.
- **Editor builders over hand-wiring.** Generated UI (`SettingsScreenBuilder` , `StatementsAppBuilder`) is produced by re-runnable menu commands that construct controls, wire every serialized reference, and save the prefab. Manual Inspector work is reserved for content, not plumbing.
- **Data-driven runtime UI.** The Statements list renders itself from `StatementLibrary` at open time; replacing filler data with real recordings requires no UI changes.
- **One theme.** `TabletTheme` is the single source of truth for colours; all generated apps read from it so the tablet looks like one operating system.
- **Component-only singleton destruction.** Persistent managers destroy just the duplicate *component* (never the `GameObject`) when a second instance appears — a duplicate manager riding on the tablet root must never delete the tablet.

2.3 The shared input seam

The **A button** serves two systems: tablet recall and gaze-door open/close. `DoorGazeManager` claims the button while the player looks at a door by setting `TabletRecallInScene.SuppressRecall` ; otherwise the press

summons the tablet. This is deliberate — one "interact" button keeps the VR control scheme minimal — but it is a coupling worth watching as more systems want input.

3. Subsystem Deep Dive

3.1 Navigation — `ScreenManager`

A show-one-screen-at-a-time switch (`ShowHome/Camera/Evidence/Notebook/Settings/Statements`). It also powers the camera rig on/off with the Camera app so the capture camera never renders in the background. New screens must be added to `HideAll()` (the newer `statementsScreen` is null-guarded; the five legacy fields are not — see §7).

3.2 Persistence & recall — `TabletPersist` , `TabletRecallInScene`

`TabletPersist` keeps the tablet alive across `CSHouse ↔ CS_Outside` ; a scene's own tablet instance self-destructs if one already persists. `TabletRecallInScene` (also persistent) rebinds the head camera and tablet after every scene load and summons the tablet in front of the player on **A**. Its spawn logic is now collision-aware:

- head forward is **flattened to the horizon** (raw forward aimed the spawn at the floor/ceiling),
- a **spherecast** clamps the spawn distance so the tablet never materialises inside a wall or door (physics depenetration used to launch it across the room),
- residual **ejection velocity is zeroed** one frame after placement,
- the main camera's **near clip plane is raised to 0.05** here as a depth-precision fix (see bug #10).

3.3 Camera & evidence — `IpadCamera` , `KeyEvidenceItem` , `EvidenceGradingManager`

A dedicated camera on the tablet renders continuously into a `RenderTexture` viewfinder. The shutter:

1. waits for **end of frame** (required on Quest/Vulkan),
2. copies the frame into an **RGBA32** `Texture2D` (mobile GPUs cannot read back RGB24),
3. runs detection: frustum test → distance gate (8 m) → **line-of-sight from the player's eyes** so the tablet can't be pushed through walls,
4. marks found evidence in `EvidenceGradingManager` and stores an `EvidencePhoto` in `PhotoLibrary` .

The occlusion ray walks **all** hits and ignores the held tablet, the player's own body/hands, and the evidence item itself — only genuine world geometry blocks. Per-item diagnostic logs name the failing gate (`too far` , `occluded`) for on-device debugging via `logcat`.

`EvidenceGradingManager` keeps a registry of `KeyEvidenceItem` s (self-registering, order-independent) and a found-set that produces a pass/fail grade. State is intentionally session-only.

3.4 Gallery — `PhotoGalleryUI`

Single-photo pagination with arrows, swipe, counter, and an **"Evidence Found" badge**. The badge is forced to bold amber and top sibling order at runtime because the authored black text vanished on night photos.

3.5 Settings — `SettingsManager` , `SettingsApplier` , `iPadSettingsPanel`

- **SettingsManager** owns the data model and persistence: one `settings.json` in `persistentDataPath`, shared by the main menu (issue #12) and the iPad (issue #13), so either surface overrides the other and preferences survive restarts. It raises `OnSettingsChanged` after every load/save.
- **SettingsApplier** turns data into world state. It re-resolves scene objects (BNG `LocomotionManager`, `SmoothLocomotion`, the sun) after every scene load, forces BNG's `PlayerPrefs` loading off, and re-applies at end-of-frame to win `Start()` ordering races. **Day** restores the scene's captured authored lighting; **night** is driven through flat dark ambient + skybox clear + sun dimming — chosen because `CSHouse` contains *no lights at all* and is lit purely by ambient.
- **iPadSettingsPanel1** wires the generated controls (teleport toggle, three speed buttons, day/night, three volume sliders) through `SettingsManager`.

Functional today: movement type, movement speed (0.75/1.0/1.25×), day/night. Saved-but-inert: the three volume channels (no audio assets/mixer yet — documented TODO).

3.6 Statements — `StatementLibrary`, `StatementsAppUI`

Lives **inside the Notebook app** behind a large centred "Recorded Statements" button. `StatementLibrary` holds entries of *either* an `AudioClip` or a `VideoClip` plus person name and date; the list always renders **ascending by parsed date** regardless of Inspector order. The app generates its whole UI at runtime: scrollable rows (`name / date · AUDIO|VIDEO · duration`), a compact bottom player bar (play/pause, scrub slider, time readout), and a 16:9 video panel that appears only during video playback. One shared `AudioSource / VideoPlayer` pair guarantees that **starting any statement overrides whatever was playing**. Filler data: four generated sweep-tone WAVs (pitch encodes position — scrubbing is instantly audible) and two real `.mov` footage clips.

3.7 Gaze doors (tablet-adjacent)

All BNG physics doors are converted at scene load: closed, frozen, every joint and BNG behaviour stripped (the handle `Grabbable` triggered a hand-scaling bug), and replaced with a gaze prompt + A-button **toggle**. Doors swing 90° *away from the player* (simulated both ways, farther end wins), open and close as exact inverses computed from the captured closed pose, and per-door components (`DoorPropAngleOverride`, `GazeDoorExclude`) provide overrides. Shares the A button with recall via the suppression flag.

3.8 Editor tooling

Tool	Menu	Produces
<code>SettingsScreenBuilder</code>	Tools ▶ Crime Scene ▶ Build iPad Settings Screen	Settings UI + wiring + managers on the root
<code>StatementsAppBuilder</code>	Tools ▶ Crime Scene ▶ Build iPad Statements App	Statements screen + library + notebook link
<code>GrabbableSetupMenu</code> / <code>GrabbableObjectSetup</code>	GameObject ▶ Crime Scene ▶ Make Grabbable	BNG grabbable configuration

Both builders are idempotent (delete-and-rebuild), refuse to run while the prefab is open in Prefab Mode, refresh the `AssetDatabase` before loading media, and log loudly when referenced assets are missing.

4. Bugs Found & Fixed

#	Symptom	Root cause	Fix
1	Settings feature "merged but missing"	Cherry-pick of PR #67 left the working UI in an orphaned scene at <code>Assets/</code> root; tablet's <code>SettingsScreen</code> was empty and unwired	Rebuilt inside the prefab via generator tool; deleted dead duplicate <code>SettingsData.cs</code>
2	(Latent) tablet would be deleted on scene entry	<code>SettingsManager</code> duplicate-check destroyed the whole <code>GameObject</code>	Destroy only the component; <code>DontDestroyOnLoad</code> only when root
3	Settings couldn't affect lights/locomotion	Persistent prefab cannot serialize scene references	<code>SettingsApplier</code> auto-resolves per scene load + end-of-frame re-apply
4	Day/night did nothing	<code>CSHouse</code> has zero lights; scenes lit by skybox ambient only	Night = flat dark ambient + skybox clear + sun dim; day = captured authored lighting, once per scene
5	Door opened a fraction of any configured angle	Rigidbody interpolation stomped per-frame transform rotation	Interpolation off + absolute-pose swing with exact final angle
6	Door prompt flickered on/off	Single closest-hit gaze ray intermittently hit the player's own body/hands/tablet	<code>RaycastAll</code> + filter player rig and tablet
7	Doors couldn't be closed	One-way design	A-button toggle; open/close are exact inverses; prompt text reflects state
8	Tablet "spawned far away" after doors/teleport	Fixed 0.6 m spawn placed tablet inside nearby geometry ; depenetration launched it	Spherecast-clamped spawn distance + post-frame velocity kill
9	Tablet spawned at feet/ceiling	Raw head forward used while looking down/up	Horizon-flattened forward, world-vertical offsets
10	Tablet UI flickering "triangles" at distance	Z-fighting: UI canvas coplanar with screen mesh + 0.01 near clip starving depth precision	Near clip raised to 0.05 on rebind
11	Photos never reached gallery on device	<code>ReadPixels</code> mid-frame + RGB24 readback — both unsupported on Quest/Vulkan	End-of-frame coroutine + RGBA32 + failure logging
12	Evidence badge invisible	Authored black text over dark night photos; z-order fragile	Forced bold amber + <code>SetAsLastSibling</code> at runtime
13	Evidence never detected in VR (badge absent)	Eye-to-item occlusion ray always hit the held tablet first	Occlusion walks all hits; ignores tablet, player, and the item itself
14	APK opened as flat black window	OpenXR Meta Quest Support feature disabled for Android; no Android interaction profiles	Enabled feature + Touch/Touch Plus profiles + validation fixes
15	"Settings icon appears twice"	Statements icon cloned the Settings icon whose sprite lives on a child — retint/relabel hit nothing	Icon built from scratch (later replaced by notebook link entirely)
16	CS0136 compile error	Local <code>button</code> shadowed a foreach variable	Rename

17	Hand grew each time a door handle was grabbed	BNG handle-grab bug	Handles are no longer grabbable at all (grab system stripped from doors)
18	Statements list appeared empty on device	Builder could load media before Unity imported it (silent nulls); stale APK	<code>AssetDatabase.Refresh()</code> first, loud error on missing media, runtime count logging

Pattern worth noting: the highest-value fixes came from *device-only* failure modes (#11, #13, #14) — the editor is too forgiving. Every capture/raycast/graphics assumption should be validated on Quest hardware before being considered done.

5. Strengths

1. **Single-prefab ownership.** The entire tablet — apps, managers, data — travels in `Tablet_manager.prefab`. Scenes need nothing; adding the tablet to a future scene is drag-and-drop.
 2. **Reproducible UI via builders.** Generated screens can be torn down and rebuilt in seconds; wiring mistakes are structurally impossible (references are assigned by code).
 3. **Auto-resolving scene integration.** Settings and doors find their per-scene targets at load; no serialized scene references to break.
 4. **Data-driven content path.** Real witness recordings replace filler by editing one component's list — no code or UI work.
 5. **Coherent theme.** One palette file; all generated surfaces agree. Restyling the tablet is a constant-change + rebuild.
 6. **Diagnostics culture.** Detection gates, capture failures, library counts, and door conversions all log with actionable messages — remote (logcat) debugging is realistic.
 7. **VR-hardened.** End-of-frame capture, RGBA32 readback, occlusion filtering, collision-aware spawning, depth-precision management: the codebase now encodes real headset lessons.
 8. **Robust persistence semantics.** One JSON file, last-writer-wins between menu and tablet, defaults on first run, corruption-tolerant load.
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6. Weaknesses

1. **The prefab lives inside the vendor folder** (`Assets/BNG Framework/Prefabs/iPad/`). A BNG upgrade could clobber it. It is *ours*, not vendor content.
 2. **Memory: photos are unbounded raw textures.** Each capture is a 1024×1024 RGBA32 `Texture2D` (~4 MB). Thirty photos ≈ 120 MB on a mobile chipset. No cap, no compression, no eviction.
 3. **Statement audio bypasses the volume settings.** No `AudioMixer` exists, so the narration slider does not affect statement playback — the most audible inconsistency in the settings system.
 4. **A-button overload.** Doors and recall share one button through a suppression flag; a third consumer would make this brittle. No haptic/visual feedback distinguishes the two actions.
 5. **Legacy null-safety gaps.** `ScreenManager.HideAll()` hard-dereferences the five original screens; a missing assignment NREs. Naming is inconsistent (`iPadSettingsPanel` vs `TabletRecallInScene` vs `IpadCamera`).
 6. **Runtime-generated UI limits art direction.** Designers cannot restyle the Statements list in the Inspector; changes require code edits (mitigated, but not removed, by `TabletTheme`).
 7. **Synchronous capture stall.** `ReadPixels` blocks the render thread; on Quest the shutter costs a visible frame hitch.
 8. **Concern leakage.** The near-clip fix lives in `TabletRecall` because it was convenient — a camera-setup concern in a recall script.
 9. **No automated tests.** Every regression so far was caught by manual play; the project has no play-mode test layer at all.
 10. `lastUpdatedFrom` **metadata lies** — every setter hardcodes `"iPad"` even when the main menu writes.
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7. Latent Issues to Fix (with suggested solutions)

Priority-ordered; none are currently player-visible blockers.

P	Issue	Suggested solution
1	Photo memory growth (\$6.2)	Cap the library (e.g. 24 photos, FIFO eviction with <code>Destroy(texture)</code>); capture at 768 ² or compress via <code>EncodeToJPG</code> + re-decode on view
1	Statement volume routing	Create the project AudioMixer (Background/Narration/SFX groups), route the statements <code>AudioSource</code> through Narration, assign the mixer on <code>SettingsApplier</code> — the plumbing already exists
1	Video transcode unverified on Android	Enable <i>Transcode</i> on the <code>.mov</code> imports and verify on device; H.264/VP8 target for Quest
2	Recall while the tablet is held	Teleporting a grabbed tablet can desync BNG's grabber; call the grabber's release before repositioning
2	Door closes through the player	A closing kinematic door sweeps through the player capsule; add an overlap check that pauses/cancels the close swing when the player stands in the arc
2	Capture frame hitch	Replace <code>ReadPixels</code> with <code>AsyncGPUReadback</code> (callback fills the <code>Texture2D</code>), keeping the end-of-frame timing
2	Vendor-folder prefab	Move <code>Tablet_manager.prefab</code> (and <code>Tablet.prefab</code> if practical) to <code>Assets/_Game/Prefabs/Tablet/</code> ; GUIDs preserve all references
3	HideAll() hard derefs	Null-guard the legacy five screens; consider a <code>List<GameObject> screens</code> refactor
3	lastUpdatedFrom accuracy	Pass a source string from the caller (menu panel vs iPad panel) into the setters
3	Near-clip fix location	Move to a small rig-setup behaviour or the <code>SettingsApplier</code> resolve step; document the 0.05 rationale where camera config lives
3	Statements UI GC churn	The list rebuilds every open; pool rows or rebuild only when the library changes
3	Day/night vs future baked lighting	If scenes ever bake lightmaps, runtime ambient stops affecting them; either keep lighting realtime or bake two lighting scenarios and swap

8. Future Directions

Near term (content readiness)

- **Real recordings pipeline.** The Statements schema is ready; define naming/date conventions with the content team and swap the filler list. Consider a `ScriptableObject` per statement if entries need per-case reuse.
- **Audio landscape.** One AudioMixer unlocks three dormant settings sliders *and* proper statement routing in a single step — highest leverage audio task.
- **Surface the grade.** `EvidenceGradingManager` computes pass/fail that nothing displays. An Evidence-app summary (found X of Y, per-item checklist) closes the core gameplay loop.
- **Subtitles & colourblind modes** (already stubbed in the settings schema) — planned as a separate issue.
- **Icon art.** The Statements link and any future apps need real sprites; the theme's generated look is a placeholder.

Mid term (systems)

- **Save system expansion.** `settings.json` proves the pattern; extend persistence to found evidence and (compressed) photos so an investigation survives app restarts — or make "session-only" an explicit design statement.
- **Statement ↔ evidence cross-linking.** Tapping a statement could highlight related evidence (and vice versa), turning two lists into an investigation web — strong detective-fantasy payoff for modest data modelling.
- **Comfort options.** Vignette on smooth locomotion, snap-turn settings — natural extensions of the existing settings schema and applier.
- **Input map consolidation.** Move the A-button sharing into a tiny interaction-priority service before a third consumer appears.

Long term (platform)

- **Case architecture.** Model a "case" (scenes, evidence set, statements, grading rules) as data so new investigations are content drops, not code forks.
 - **Automated testing.** Play-mode tests for the pure-logic cores (grading registry, settings round-trip, statement sorting) plus a device smoke checklist; CI via GameCI when the team is ready.
 - **Performance budget.** Establish a Quest 3S frame budget; the tablet's world-space canvas, continuous viewfinder rendering, and photo memory are the first line items to instrument.
 - **Localization** of all generated UI text (the builders centralise strings, which will help).
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9. Appendix

9.1 File map (tablet-related, `Assets/_Game/Scripts/`)

Area	Files
Navigation	<code>Tablet/ScreenManager.cs</code> , <code>Tablet/TabletTheme.cs</code>
Persistence/recall	<code>SceneFlow/TabletPersist.cs</code> , <code>Tablet/TabletRecall.cs</code>
Camera/evidence	<code>Camera/IpadCamera.cs</code> , <code>Camera/KeyEvidenceItem.cs</code> , <code>Camera/EvidenceGradingManager.cs</code> , <code>Camera/PhotoLibrary.cs</code> , <code>Camera/EvidencePhoto.cs</code> , <code>Camera/PhotoGalleryUI.cs</code>
Settings	<code>Settings/SettingsManager.cs</code> , <code>Settings/SettingsApplier.cs</code> , <code>Settings/iPadSettingsPanel.cs</code> , <code>Settings/Editor/SettingsScreenBuilder.cs</code>
Statements	<code>Statements/StatementLibrary.cs</code> , <code>Statements/StatementsAppUI.cs</code> , <code>Statements/Editor/StatementsAppBuilder.cs</code>
Doors (shared input)	<code>Core/GazeDoorSystem.cs</code> , <code>Core/DoorPropAngleOverride.cs</code> , <code>Core/GazeDoorExclude.cs</code>

9.2 Manual test checklist (device)

1. Launch APK → immersive VR, controllers tracked.
2. Recall tablet near a wall / after teleport → spawns in reach, never launches away.
3. Photograph evidence at ~2 m through the tablet → gallery photo + amber badge + grading log.
4. Night mode → both scenes darken; day restores authored look; persists across restart.
5. Teleport/continuous toggle + speed buttons behave; `settings.json` updates.
6. Notebook → Recorded Statements → six entries oldest-first; audio scrub audible by pitch; video panel plays with sound; new selection overrides old; audio stops on leaving the app.
7. Doors: prompt steady, opens away from player, closes flush, excluded doors (if any) still grabbable.

Report generated from the working tree on branch `TabLetCameraAndGaLLeryAppsUpdate` ; reflects code state as of 2026-07-07.